

Dyadic Haar Block Decay for Smooth Markov Kernels

The Analytic Origin of the Quarter–Half Law in Nested Physical Transfer Matrices

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Abstract

Two preceding computer-assisted papers found an unexpectedly precise scaling law under dyadic grid refinement. In unnormalized Haar coordinates, the coarse-consistency and detail blocks of the stored physical two-step matrix shrink by approximately one quarter, while the two cross channels shrink by approximately one half. We prove that this pattern is forced by smooth-kernel cancellation rather than by a numerical accident.

Let $k \in C^2([0, 1]^2)$ and let the midpoint Nyström matrix at coarse mesh h have entries $hk(x_i, y_j)$. Exact replication/detail coordinates give four blocks E_h, C_h, B_h, D_h . If $M_x, M_y, M_{xx}, M_{xy}, M_{yy}$ are uniform derivative bounds, then

$$\|E_h\|_2 \leq \frac{h^2}{32}(M_{xx} + 2M_{xy} + M_{yy}), \quad \|D_h\|_2 \leq \frac{h^2}{16}M_{xy},$$

and

$$\|C_h\|_2 \leq \frac{h}{4}M_x, \quad \|B_h\|_2 \leq \frac{h}{4}M_y.$$

We show that these rates survive discrete row normalization of a smooth positive raw kernel, subtraction of a smooth finite-rank projector, and exact operator squaring. Hence fixed-noise folded Gaussian kernels possess the quarter–half law at the analytic level.

The rigorous stored-binary64 block uppers from the $2048 \rightarrow 4096$ and $4096 \rightarrow 8192$ physical refinements become nearly identical after division by h^2, h, h, h^2 : their largest relative spread is 3.18×10^{-4} . A componentwise floating decomposition shows the same law separately for the Markov matrix, peripheral projector, extracted one-step matrix, and physical two-step matrix. The remaining gap to an all-grid theorem is now explicit: uniform control of the hard Gaussian cutoff, validated convergence of the computed peripheral projectors, and the hierarchical nonnormal resolvent recursion.

1 Introduction

The nested-grid certificates in Wang [5, 6] proved the finite stored-matrix count chain

$$N_\Gamma(A_{2048}) = N_\Gamma(A_{4096}) = N_\Gamma(A_{8192}) = 1$$

at fixed Gaussian width $\sigma = 10^{-2}$. Their exact dyadic decompositions also exposed a striking regularity. From the first refinement to the second, the four physical Haar-block bounds changed by

$$0.249921, \quad 0.500046, \quad 0.500031, \quad 0.250024.$$

The order is coarse consistency, coarse-to-detail, detail-to-coarse, and detail. This paper explains those numbers analytically.

The mechanism is the elementary vanishing-moment structure of the Haar split [2]. Averaging two fine source points and two fine target points cancels first derivatives, leaving a second-order

consistency error. Differencing in one variable leaves one derivative and hence a first-order cross channel. Differencing in both variables leaves a mixed second derivative and hence a second-order detail block. Because the coordinate transform is a scalar multiple of an orthogonal matrix, these cancellations pass directly to Euclidean operator norms.

There are three reasons the conclusion is not automatic for the stored physical matrices.

1. The Gaussian matrix is normalized row by row using a discrete midpoint denominator, not the continuum normalizer.
2. The Perron and parity modes are subtracted through a dimension-dependent biorthogonal rank-two projector.
3. The physical object is the square of the extracted one-step matrix.

We prove that each operation preserves the first/second-order Haar pattern under natural smoothness and approximation hypotheses. The full row-normalized folded Gaussian kernel satisfies the smoothness assumptions at every fixed positive noise width. The theorem therefore identifies the analytic route to a uniform nested-grid theory.

The result is deliberately separated into three evidence levels.

Analytic theorem

Exact inequalities for arbitrary C^2 kernels, discrete normalization, smooth finite-rank subtraction, and block squaring.

Rigorous stored ledger

Previously certified binary64 physical block uppers, rescaled by the powers predicted by the theorem.

Floating mechanism audit

Singular-value decompositions of the Markov, projector, one-step, and two-step components, used only to localize where the law appears.

The paper does not claim a uniform theorem for the exact sparse stored family. The eight-sigma hard cutoff and the computed peripheral projectors require additional validated estimates. Nor does block decay by itself control the nonnormal resolvent recursion needed for an infinite count continuation.

2 Midpoint matrices and exact Haar coordinates

Let $n \geq 1$, $h = 1/n$, and

$$x_i = y_i = \left(i + \frac{1}{2}\right) h, \quad 0 \leq i < n.$$

For a kernel $k : [0, 1]^2 \rightarrow \mathbb{C}$, define the midpoint Nyström matrix

$$(\mathbf{K}_h)_{ij} = h k(x_i, y_j). \tag{1}$$

The fine mesh is $h/2$. Within coarse cell i , its two fine midpoints are

$$x_i^- = x_i - \frac{h}{4}, \quad x_i^+ = x_i + \frac{h}{4},$$

and similarly for $y_j^\pm$.

Define replication and alternating-detail injections $J, W : \mathbb{C}^n \rightarrow \mathbb{C}^{2n}$ by

$$(Jv)_{2i} = (Jv)_{2i+1} = v_i,$$

$$(Wv)_{2i} = v_i, \quad (Wv)_{2i+1} = -v_i.$$

Their restrictions are

$$R = \frac{1}{2}J^*, \quad S = \frac{1}{2}W^*.$$

The exact transform and inverse are

$$T = \begin{bmatrix} J & W \end{bmatrix}, \quad T^{-1} = \begin{bmatrix} R \\ S \end{bmatrix}.$$

Since $T = \sqrt{2}Q$ for a unitary Q ,

$$T^{-1}MT = Q^*MQ, \quad \|T^{-1}MT\|_2 = \|M\|_2. \quad (2)$$

The four fine-grid Haar blocks relative to K_h are

$$E_h = RK_{h/2}J - K_h, \quad C_h = SK_{h/2}J, \quad (3)$$

$$B_h = RK_{h/2}W, \quad D_h = SK_{h/2}W. \quad (4)$$

Thus

$$T^{-1}K_{h/2}T = \begin{pmatrix} K_h + E_h & B_h \\ C_h & D_h \end{pmatrix}. \quad (5)$$

Our naming follows the direction of propagation: C_h maps coarse input to detail output, while B_h maps detail input to coarse output.

3 The smooth-kernel quarter-half theorem

For $k \in C^2([0, 1]^2)$, set

$$M_x = \|\partial_x k\|_\infty, \quad M_y = \|\partial_y k\|_\infty, \quad M_{xx} = \|\partial_{xx} k\|_\infty,$$

$$M_{xy} = \|\partial_{xy} k\|_\infty, \quad M_{yy} = \|\partial_{yy} k\|_\infty.$$

Theorem 3.1 (Dyadic Haar block decay). *The exact blocks in (3)–(4) satisfy*

$$\|E_h\|_2 \leq \frac{h^2}{32}(M_{xx} + 2M_{xy} + M_{yy}), \quad (6)$$

$$\|C_h\|_2 \leq \frac{h}{4}M_x, \quad (7)$$

$$\|B_h\|_2 \leq \frac{h}{4}M_y, \quad (8)$$

$$\|D_h\|_2 \leq \frac{h^2}{16}M_{xy}. \quad (9)$$

Proof. The coarse-consistency entries are

$$(E_h)_{ij} = h \left\{ \frac{1}{4} \sum_{a,b \in \{-,+\}} k(x_i^a, y_j^b) - k(x_i, y_j) \right\}. \quad (10)$$

The symmetric average cancels the two first-order Taylor terms. Since every displacement has modulus $h/4$, the second-order integral remainder gives

$$|(E_h)_{ij}| \leq \frac{h^3}{32} (M_{xx} + 2M_{xy} + M_{yy}).$$

An $n \times n$ matrix whose entries have modulus at most a has 1- and ∞ -norm at most na , hence spectral norm at most na . Using $nh = 1$ proves (6).

For the coarse-to-detail block,

$$(C_h)_{ij} = \frac{h}{4} \sum_{b \in \{-,+\}} \{k(x_i^-, y_j^b) - k(x_i^+, y_j^b)\}. \quad (11)$$

Each source difference is bounded by $(h/2)M_x$, so $|(C_h)_{ij}| \leq h^2 M_x/4$. The same matrix-norm argument proves (7). Interchanging source and target variables proves (8).

Finally,

$$(D_h)_{ij} = \frac{h}{4} \{k(x_i^-, y_j^-) - k(x_i^-, y_j^+) - k(x_i^+, y_j^-) + k(x_i^+, y_j^+)\}. \quad (12)$$

The bracket is a mixed rectangular difference. By two applications of the fundamental theorem of calculus, its modulus is at most $(h/2)^2 M_{xy}$. Thus $|(D_h)_{ij}| \leq h^3 M_{xy}/16$, which proves (9). $\square$

Corollary 3.2 (Quarter-half law). *If the derivative envelope is held fixed while the coarse mesh changes from h to $h/2$, the theorem's consistency and detail uppers are divided by four, while its two cross-channel uppers are divided by two.*

The theorem is deterministic and finite-dimensional. It does not invoke an asymptotic spectral approximation result. The only ingredients are midpoint geometry, one Haar vanishing moment, and the Euclidean Schur estimate.

4 Discrete row normalization preserves the rates

The stored Markov matrices are normalized by a discrete denominator. We now show that this quotient operation contributes only a second-order defect.

Let $g \in C^2([0, 1]^2)$ be positive and define

$$Z(x) = \int_0^1 g(x, y) dy, \quad z_* := \inf_{x \in [0, 1]} Z(x) > 0,$$

$$k(x, y) = \frac{g(x, y)}{Z(x)}.$$

The continuum-normalized midpoint matrix is $\mathbf{K}_h$ from (1). The separately normalized discrete matrix is

$$(\widehat{\mathbf{K}}_h)_{ij} = \frac{g(x_i, y_j)}{\sum_{\ell=0}^{n-1} g(x_i, y_\ell)} = h \frac{g(x_i, y_j)}{Z_h(x_i)}, \quad (13)$$

where

$$Z_h(x) = h \sum_{j=0}^{n-1} g(x, y_j).$$

Put

$$G_0 = \|g\|_\infty, \quad G_{yy} = \|\partial_{yy}g\|_\infty, \quad \eta_h = \frac{h^2}{24}G_{yy}.$$

Composite midpoint quadrature gives

$$\sup_x |Z_h(x) - Z(x)| \leq \eta_h. \quad (14)$$

Proposition 4.1 (Discrete normalization defect). *Assume $\eta_h < z_*$. Then*

$$\|\hat{K}_h - K_h\|_2 \leq q_h := \frac{G_0 \eta_h}{z_*(z_* - \eta_h)}. \quad (15)$$

If $\hat{E}_h, \hat{C}_h, \hat{B}_h, \hat{D}_h$ are the Haar blocks of the discretely normalized matrices, then

$$\|\hat{E}_h\|_2 \leq \frac{h^2}{32}(M_{xx} + 2M_{xy} + M_{yy}) + q_h + q_{h/2}, \quad (16)$$

$$\|\hat{C}_h\|_2 \leq \frac{h}{4}M_x + q_{h/2}, \quad (17)$$

$$\|\hat{B}_h\|_2 \leq \frac{h}{4}M_y + q_{h/2}, \quad (18)$$

$$\|\hat{D}_h\|_2 \leq \frac{h^2}{16}M_{xy} + q_{h/2}. \quad (19)$$

Proof. By (14), every entry difference is bounded by

$$hG_0 \frac{\eta_h}{z_*(z_* - \eta_h)}.$$

Multiplication by $n = 1/h$ bounds both the 1- and ∞ -norms by q_h , proving (15). The coordinate products satisfy

$$\|R\|_2 \|J\|_2 = \|R\|_2 \|W\|_2 = \|S\|_2 \|J\|_2 = \|S\|_2 \|W\|_2 = 1.$$

Therefore the coarse defect contributes at most q_h , the fine defect at most $q_{h/2}$, and the stated estimates follow from [theorem 3.1](#). $\square$

Since $q_h = O(h^2)$, row normalization preserves all four rates.

4.1 Folded Gaussian corollary

For fixed $\sigma > 0$ and a smooth map f , consider the folded raw kernel

$$g_\sigma(x, y) = \phi_\sigma(y - f(x)) + \phi_\sigma(y + f(x)), \quad (20)$$

where

$$\phi_\sigma(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-t^2/(2\sigma^2)}.$$

On the compact square, g_σ is smooth and its row integral is strictly positive. Hence its normalized kernel is C^∞ .

Corollary 4.2 (Fixed-noise folded Gaussian rate). *At every fixed $\sigma > 0$, the full discretely row-normalized folded Gaussian midpoint matrices satisfy*

$$\|E_h\|_2, \|D_h\|_2 = O(h^2), \quad \|C_h\|_2, \|B_h\|_2 = O(h).$$

This corollary concerns the full Gaussian sum. The stored matrices retain a hard eight-sigma support window. Its omitted tail is tiny at the archived scales, but the indicator boundary is not a C^2 kernel. A uniform theorem for the exact sparse family must add an explicit cutoff-defect estimate.

5 Smooth extraction and exact squaring

We next show why peripheral subtraction and two-step composition do not change the powers of h .

5.1 Finite-rank smooth projectors

Let

$$q(x, y) = \sum_{a=1}^r \lambda_a r_a(x) \ell_a(y) \tag{21}$$

with $r_a, \ell_a \in C^2[0, 1]$, and set $u = k - q$. Since $u \in C^2$, the midpoint matrices of u obey [theorem 3.1](#) directly.

The computed finite-dimensional projector need not be an exact sample of q . The following stability statement isolates the required approximation.

Proposition 5.1 (Projector approximation stability). *Let Q_h be the midpoint matrix of (21), and suppose stored projectors satisfy*

$$\|\hat{Q}_h - Q_h\|_2 \leq \kappa h^2.$$

Then subtracting $\hat{Q}_h$ from a matrix family satisfying (16)–(19) preserves the rates

$$E, D = O(h^2), \quad C, B = O(h).$$

More precisely, the projector defect adds at most $5\kappa h^2/4$ to the coarse-consistency upper and at most $\kappa h^2/4$ to each fine-only block.

Proof. Write the stored projector error as F_h . The coarse-consistency error is

$$RF_{h/2}J - F_h,$$

whose norm is at most

$$\kappa(h/2)^2 + \kappa h^2 = \frac{5}{4}\kappa h^2.$$

Each other block contains only a coordinate compression of $F_{h/2}$, bounded by $\kappa h^2/4$. $\square$

For isolated smooth continuum eigenmodes, standard collectively compact Nyström theory suggests such projector convergence [1, 3, 4]. The present archive does not turn that suggestion into a validated projector enclosure for the parity mode; this is one of the remaining gates.

5.2 Squaring preserves the powers

Suppose an extracted one-step family has exact fine-grid block form

$$T^{-1}U_{h/2}T = \begin{pmatrix} U_h + E_h & B_h \\ C_h & D_h \end{pmatrix}. \quad (22)$$

Let $A_h = U_h^2$. Squaring (22) gives

$$T^{-1}A_{h/2}T = \begin{pmatrix} A_h + U_h E_h + E_h U_h + E_h^2 + B_h C_h & (U_h + E_h)B_h + B_h D_h \\ C_h(U_h + E_h) + D_h C_h & C_h B_h + D_h^2 \end{pmatrix}. \quad (23)$$

Theorem 5.2 (Closure under operator squaring). *Assume, for $0 < h \leq h_0$,*

$$\|U_h\|_2 \leq M, \quad \|E_h\|_2 \leq eh^2, \quad \|C_h\|_2 \leq ch, \quad \|B_h\|_2 \leq bh, \quad \|D_h\|_2 \leq dh^2.$$

Then the physical two-step blocks satisfy

$$\|E_h^{(2)}\|_2 \leq (2Me + bc + e^2 h_0^2)h^2, \quad (24)$$

$$\|C_h^{(2)}\|_2 \leq c\{M + (e + d)h_0^2\}h, \quad (25)$$

$$\|B_h^{(2)}\|_2 \leq b\{M + (e + d)h_0^2\}h, \quad (26)$$

$$\|D_h^{(2)}\|_2 \leq (bc + d^2 h_0^2)h^2. \quad (27)$$

Proof. Apply the triangle inequality and submultiplicativity to the four entries in (23). For example,

$$\|U_h E_h + E_h U_h + E_h^2 + B_h C_h\|_2 \leq 2Meh^2 + e^2 h^4 + bch^2.$$

The remaining three estimates are identical one-line calculations. $\square$

Thus the quarter-half law passes from a smooth extracted one-step kernel to the physical two-step matrix used in the count certificates.

6 Stored-model scaling ledger

The analytic theorem concerns exact smooth-kernel families. We now compare it with the two rigorous stored physical refinements. No new interval enclosure is substituted for the inherited certificates: the block uppers and their hashes are replayed exactly from RH-36 and RH-37.

6.1 Componentwise mechanism audit

The stored one-step matrix is

$$U = P - Q,$$

where P is the row-normalized sparse Markov matrix and Q is the rank-two Perron/parity term. The physical matrix is $A = U^2$. Floating leading singular values were computed for every Haar block of P, Q, U, A at both refinements. Their second-to-first ratios are shown in [table 1](#).

Table 1: Floating component ratios. These identify the mechanism but are not used as rigorous inequalities.

component	E	C	B	D
Markov P	0.250055	0.500086	0.500026	0.250067
projector Q	0.249780	0.500028	0.500052	0.250029
one-step $U = P - Q$	0.250055	0.500086	0.500024	0.250067
two-step $A = U^2$	0.249920	0.500046	0.500031	0.250024

The maximum deviation from the exact target values $1/4$ and $1/2$ is 2.203×10^{-4} . The corresponding effective exponents $-\log_2(\text{ratio})$ differ from $2, 1, 1, 2$ by less than 4.60×10^{-4} . In particular, peripheral subtraction is not the source of the law: P and Q each possess it separately.

6.2 Rigorous physical renormalization

Let the coarse mesh at a refinement be h . Divide the rigorous physical uppers for E, D by h^2 and those for C, B by h . The resulting ledger is [table 2](#).

Table 2: Rigorous stored-binary64 physical uppers after division by the powers predicted in [theorems 3.1](#) and [5.2](#).

block	power	2048 $\rightarrow$ 4096	4096 $\rightarrow$ 8192
E	h^2	834.1513371	833.8864287
C	h	23.1863048	23.1884515
B	h	28.1039701	28.1057350
D	h^2	547.4492687	547.5026097

These are ratios and rescalings of rigorous upper bounds, not lower enclosures for the true block norms. Nevertheless, they form an exact reproducible property of the two certificate ledgers. The largest max/min spread between the two renormalized uppers is

$$1.0003176793,$$

or about 0.032%.

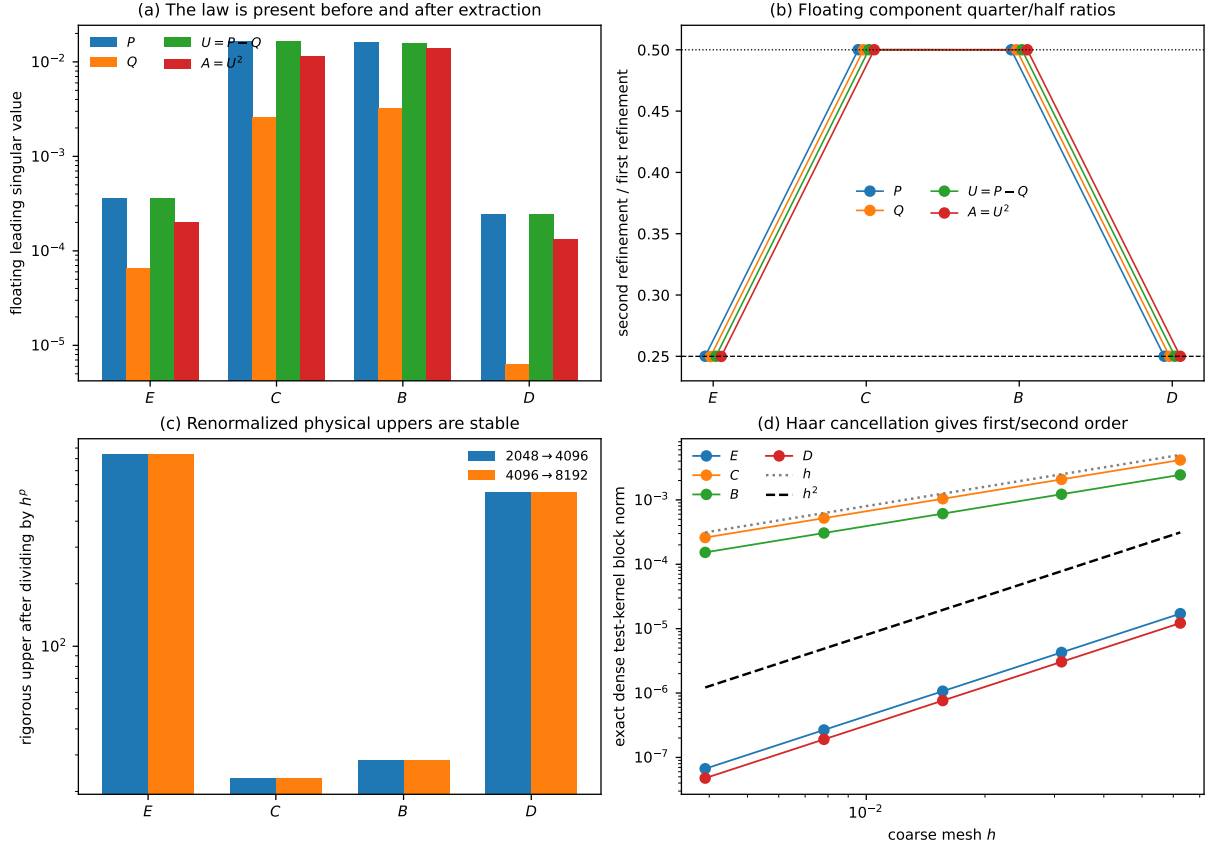


Figure 1: Analytic mechanism and finite-scale evidence. (a) Floating first refinement block norms for the Markov, peripheral, extracted one-step, and physical two-step components. (b) All four component families follow the quarter–half ratios independently. (c) The rigorous physical uppers become nearly level-independent after division by h^2, h, h, h^2 . (d) An exact dense smooth test kernel illustrates the first- and second-order slopes proved in [theorem 3.1](#).

7 What is now proved, and what remains

The analytic conclusion is stronger than the original two-point numerical observation:

$$\boxed{\text{one Haar difference} = O(h), \quad \text{two Haar differences} = O(h^2)}. \quad (28)$$

It survives the three operations that define the intended physical family: row normalization, smooth finite-rank extraction, and squaring. Consequently, if the exact stored sparse/projector family can be placed under the hypotheses uniformly, its Schur self-energy obeys

$$\|E_h\|_2 + \|B_h\|_2 \left\| (z - D_h)^{-1} \right\|_2 \|C_h\|_2 = O(h^2)$$

on any contour whose distance from the vanishing detail spectrum is bounded below.

This is precisely the perturbative scaling needed by the nested count route. It is not yet sufficient for an infinite refinement theorem. Three gates remain.

1. **Hard-cutoff defect.** The eight-sigma sparse support must be compared uniformly with the full smooth Gaussian matrix, or the cutoff must be allowed to grow with refinement.
2. **Peripheral projector enclosure.** The stored Perron/parity projectors must be enclosed relative to a smooth isolated continuum spectral projector with an $O(h^2)$ operator bound.

3. **Hierarchical resolvent recursion.** Even when the effective perturbation is $O(h^2)$, nonnormal resolvent bounds propagated through successive Schur inverses can grow. Their scalar recursion needs a separate summability or bootstrap theorem.

The first two gates connect the exact stored family to the smooth-kernel theorem. The third determines whether block decay can be converted into an all-grid contour-count stabilization result.

8 Archive and reproducibility

The machine-readable certificate records:

- the exact formulas implemented for ideal Haar bounds, discrete normalization defects, block squaring, and renormalized constants;
- hashes of the inherited RH-36 and RH-37 rigorous block certificates;
- the four rigorous physical scaled ledgers and their spreads;
- all floating component ratios and effective exponents; and
- an explicit list of the theorem’s scope limitations.

The source tests include exact dense polynomial-kernel checks of all four Haar inequalities, a quotient-normalization check, and a random dense block-square check. The fast replay is

```
python -m pytest -q -p no:cacheprovider
python experiments/build_decay_certificate.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
cp main.pdf dyadic-haar-block-decay.pdf
python experiments/build_archive.py
python experiments/verify_archive.py
```

The theorem concerns fixed positive noise and smooth full kernels. It does not prove the exact sparse all-grid limit, a zero-noise limit, a zeta-zero identification, a Hilbert–Pólya construction, or the Riemann hypothesis.

9 Conclusion

The quarter-half law seen in two large stored refinements has a simple and rigorous origin: one Haar difference extracts one derivative, while two Haar differences extract a mixed second derivative. Midpoint consistency and the pure detail block are therefore second order; the cross channels are first order. Discrete row normalization, smooth spectral extraction, and operator squaring preserve those powers.

The physical certificate ledgers agree with the theorem to high precision after the predicted renormalization, and the component decomposition shows that the law is already present in both the Gaussian Markov kernel and the peripheral projector. The maze is therefore narrower: the next obstruction is no longer the block scaling itself, but a uniform cutoff/projector bridge and control of the resulting multilevel resolvent recursion.

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